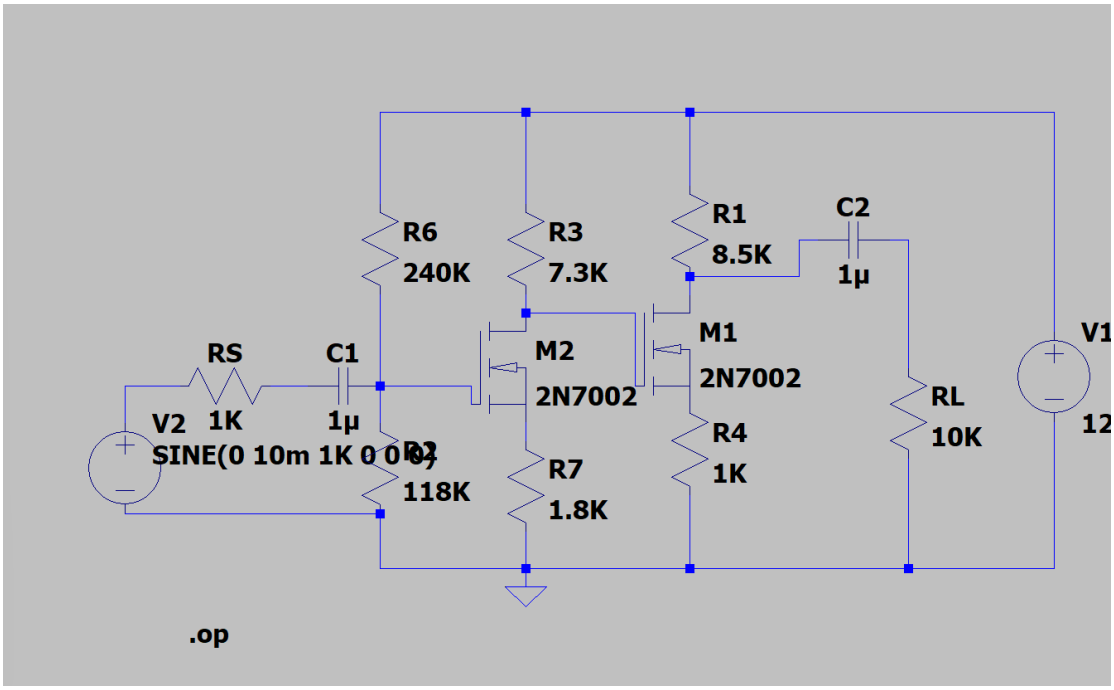
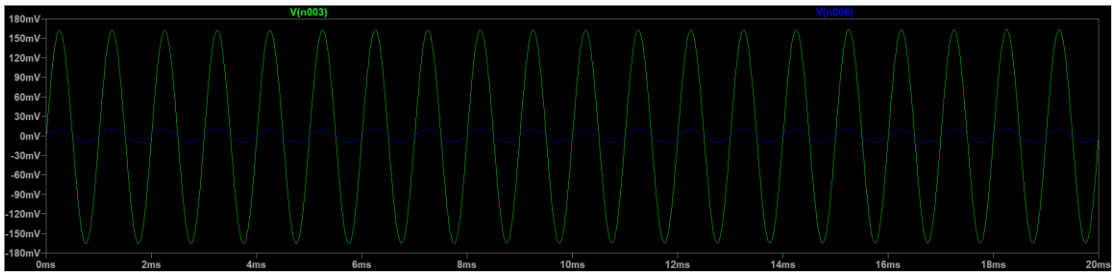
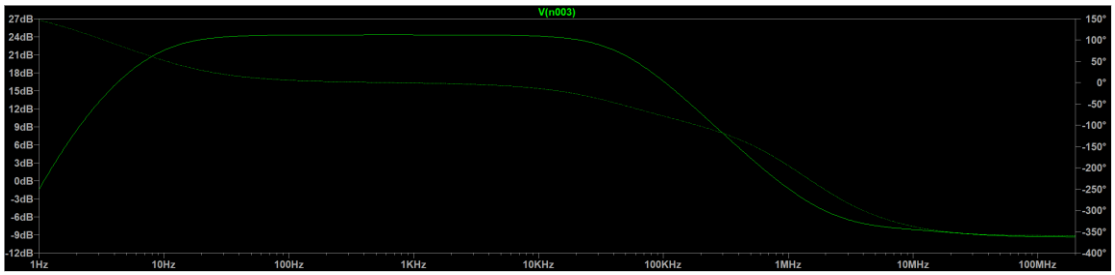


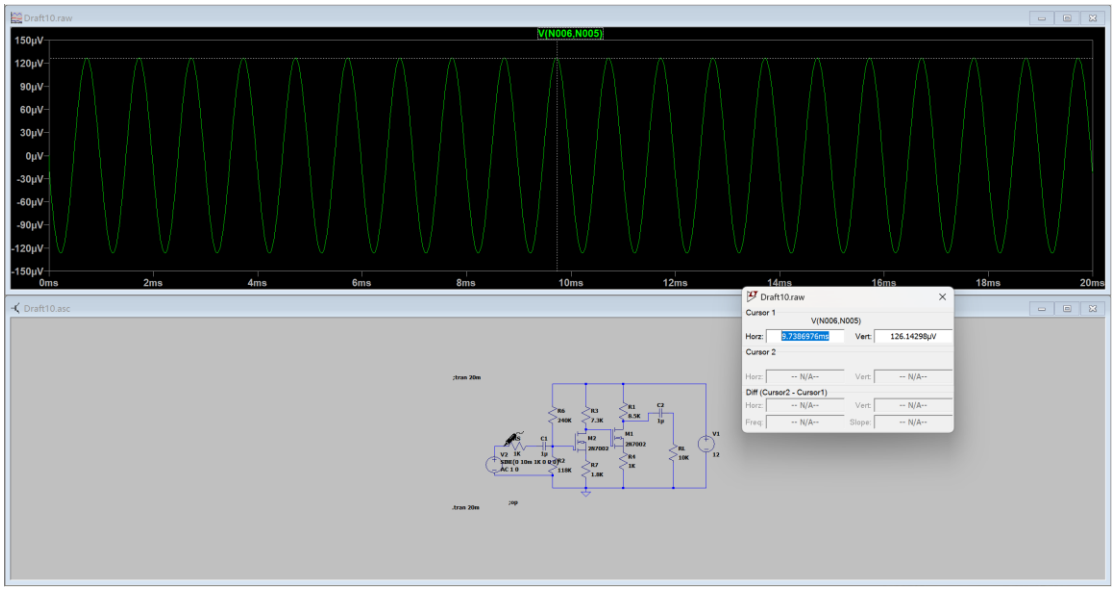
* C:\Users\28100\Documents\LTspiceXVII\Draft10.asc

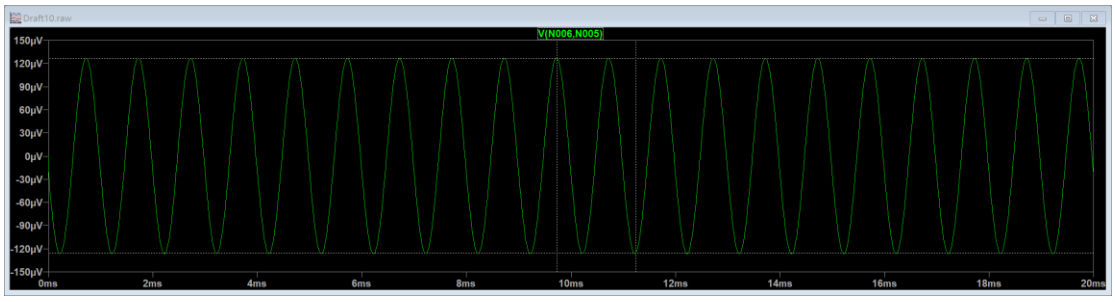
```
       --- Operating Point ---

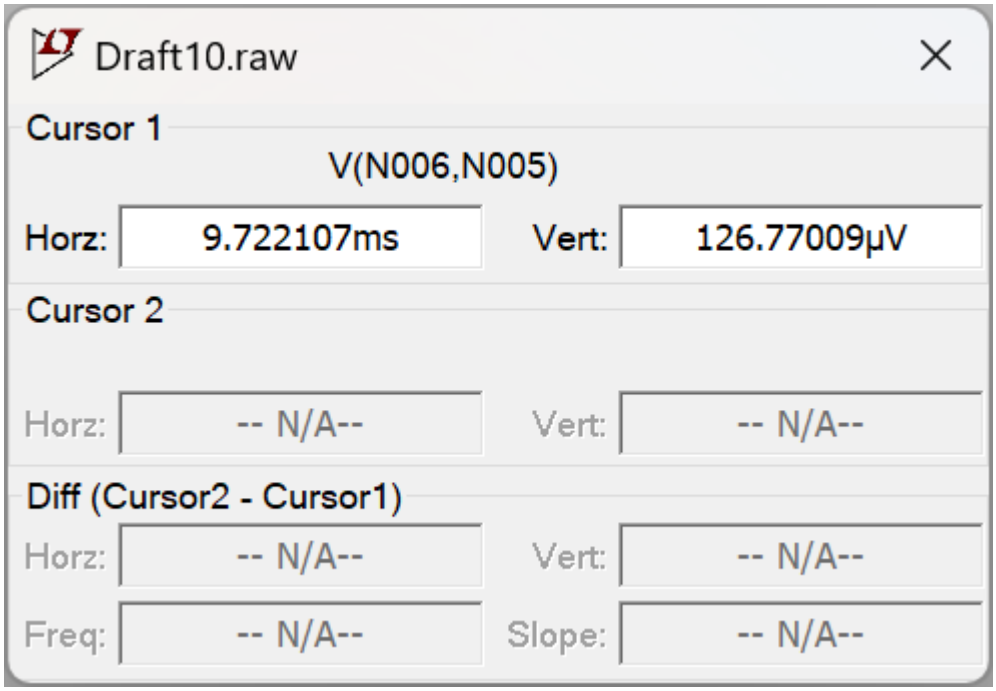
V(n005):        0              voltage
V(n006):        3.95531e-015   voltage
V(n007):        3.95531        voltage
V(n001):        12             voltage
V(n004):        2.80316        voltage
V(n008):        2.26771        voltage
V(n002):        2.43494        voltage
V(p001):        1.1253         voltage
V(n003):        2.43494e-014   voltage
Id(M1):         0.0011253      device_current
Ig(M1):         -6.28713e-011                 device_current
Is(M1):         -0.0011253     device_current
Id(M2):         0.00125984     device_current
Ig(M2):         -7.51982e-011                 device_current
Is(M2):         -0.00125984    device_current
I(C2):          -2.43494e-018                 device_current
I(C1):          3.95531e-018   device_current
I(R1):          2.43494e-018   device_current
I(R4):          0.0011253      device_current
I(R1):          0.0011253      device_current
I(R7):          0.00125984     device_current
I(R3):          0.00125984     device_current
I(R6):          3.35196e-005   device_current
I(R2):          3.35196e-005   device_current
I(Rs):          3.95531e-018   device_current
I(V1):          -0.00241866    device_current
I(V2):          3.95531e-018   device_current
```

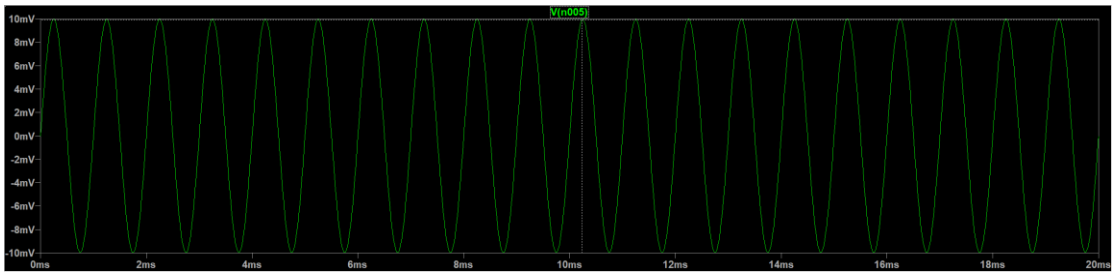

V(n003)
V(n006)
180mV
150mV
120mV
90mV
60mV
30mV
0mV
-30mV
-60mV
-90mV
-120mV
-150mV
-180mV
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
16ms
18ms
20ms

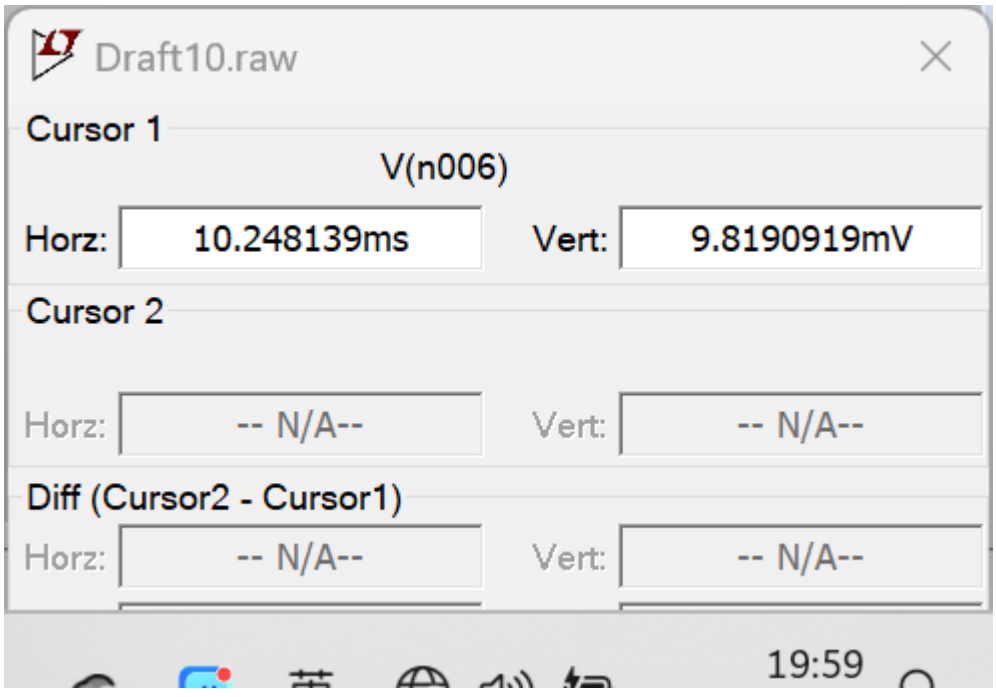

V(n003)
27dB
24dB
21dB
18dB
15dB
12dB
9dB
6dB
3dB
0dB
-3dB
-6dB
-9dB
-12dB
150°
100°
50°
0°
-50°
-100°
-150°
-200°
-250°
-300°
-350°
-400°
1Hz
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
10MHz
100MHz


Draft10.raw
V(N006,N005)
150µV
120µV
90µV
60µV
30µV
0µV
-30µV
-60µV
-90µV
-120µV
-150µV
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
16ms
18ms
20ms
Draft10.asc
Draft10.raw
Cursor 1
V(N006,N005)
Horz: 9.7386976ms
Vert: 126.14298µV
Cursor 2
Horz: -- N/A--
Vert: -- N/A--
Diff (Cursor2 - Cursor1)
Horz: -- N/A--
Vert: -- N/A--
Freq: -- N/A--
Slope: -- N/A--

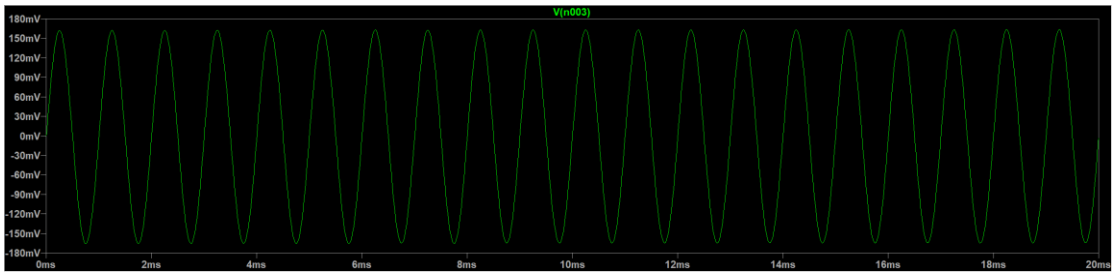

Draft10.raw
V(N006,N005)
150µV
120µV
90µV
60µV
30µV
0µV
-30µV
-60µV
-90µV
-120µV
-150µV
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
16ms
18ms
20ms

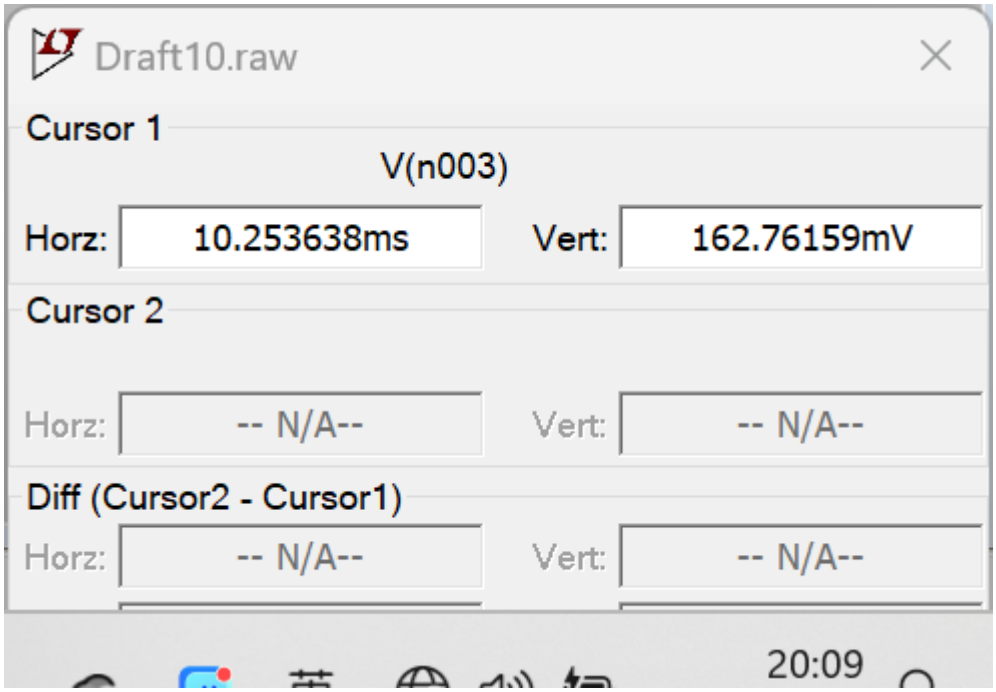


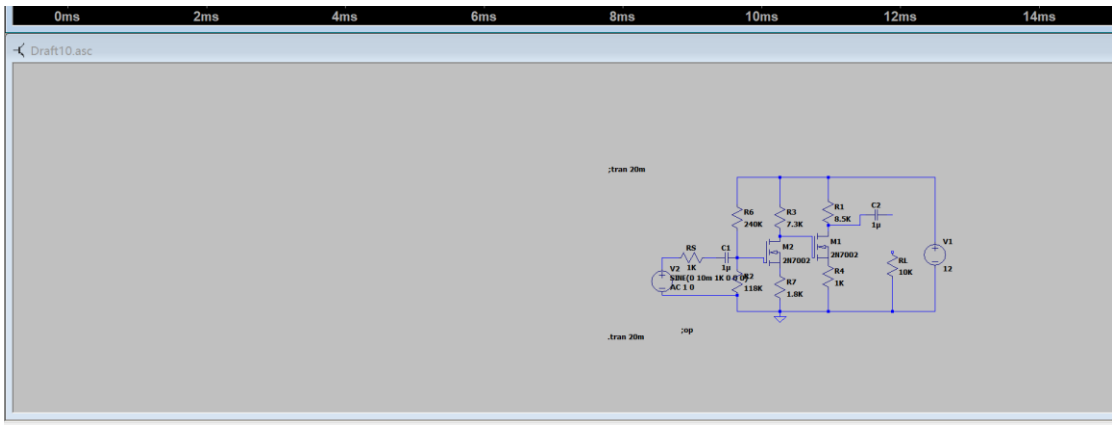


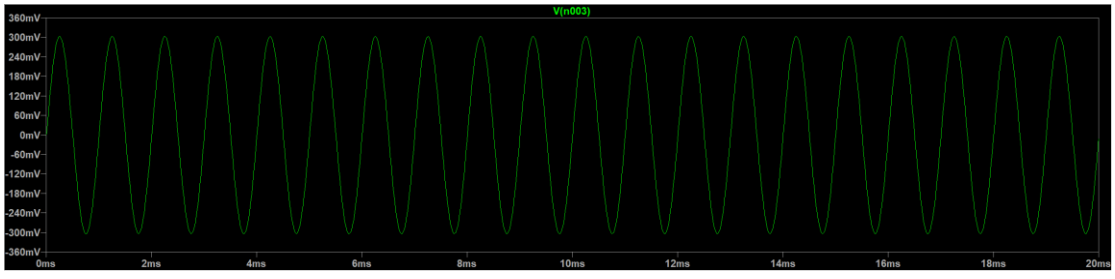


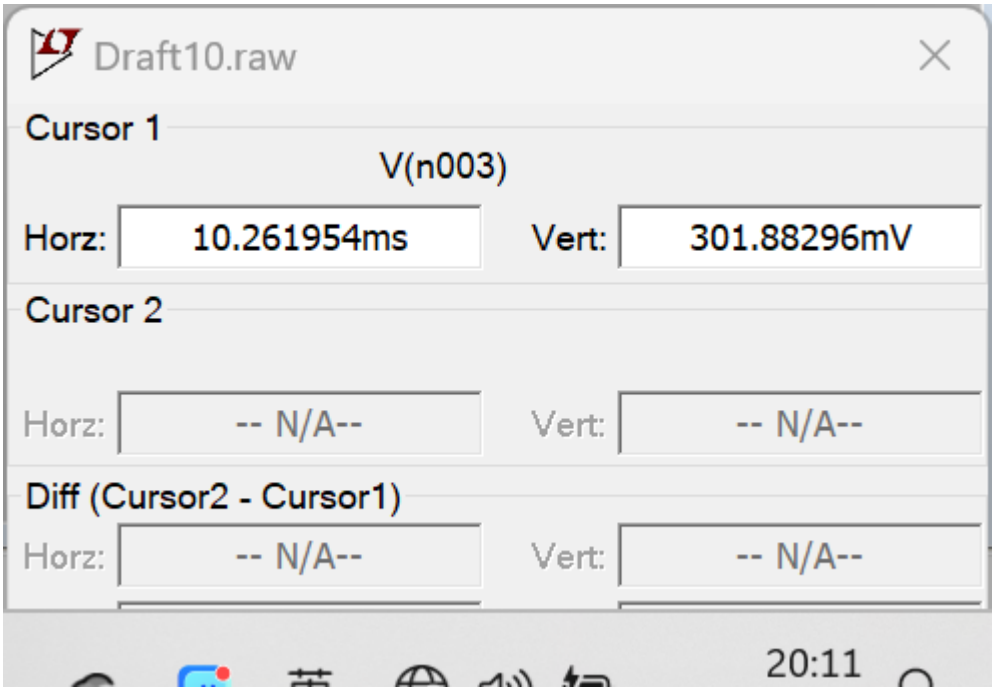
$R_i=\frac{V_i}{I_i}=\frac{V_i}{V_s-V_i}R_s$ =77384Ω


V(n003)
180mV
150mV
120mV
90mV
60mV
30mV
0mV
-30mV
-60mV
-90mV
-120mV
-150mV
-180mV
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
16ms
18ms
20ms


Draft10.raw
Cursor 1
V(n003)
Horz: 10.253638ms
Vert: 162.76159mV
Cursor 2
Horz: -- N/A--
Vert: -- N/A--
Diff (Cursor2 - Cursor1)
Horz: -- N/A--
Vert: -- N/A--
20:09


0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
Draft10.asc
;tran 20m
R6
240K
R3
7.3K
R1
8.5K
C2
1µ
M1
2N7002
M2
2N7002
R5
1K
C1
1µ
V2
1K
SINE(0 10m 1K 0 0 0)
AC 1 0
R7
1.8K
R4
1K
RL
10K
V1
12
.tran 20m
;op


V(n003)
360mV
300mV
240mV
180mV
120mV
60mV
0mV
-60mV
-120mV
-180mV
-240mV
-300mV
-360mV
0ms
2ms
4ms
6ms
8ms
10ms
12ms
14ms
16ms
18ms
20ms



$$R_o = (\frac{V_o}{V_L} - 1)R_L$$

=8.549KΩ